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Enhancing Seismic Data Augmentation with Geostatistical Features and Internal Multiples for Improved Deep Learning Applications

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Abstract

Seismic Data Augmentation (SDA) is a comprehensive framework designed to generate realistic synthetic seismic datasets for deep learning applications in seismic interpretation. It addresses the limitations of conventional data acquisition by producing structurally and geologically meaningful data that reflect the complexity and variability of real subsurface conditions. We present an advanced SDA workflow that synthesizes realistic seismic datasets through the integration of three key components: structural interpolation, stochastic simulation, and wavefield modeling. First, Image-Guided Interpolation (IGI) is applied to construct continuous velocity models from sparse well log inputs while preserving large-scale structural trends, with the option to incorporate faults and horizons if provided as input. Second, Sequential Gaussian Simulation (SGS) introduces geostatistical perturbations at smaller scales, enriching the models with spatial variability consistent with subsurface heterogeneity. Third, reflectivity-based modeling is used to generate synthetic seismic data that can include primary reflections, internal multiples, or their combination, offering flexibility in simulating realistic wavefield responses. The SDA framework is demonstrated on a public field dataset to illustrate its ability to generate structurally coherent, statistically diverse, and physically plausible seismic volumes. the resulting augmented data provide a strong foundation for both qualitative interpretation and future ML-based workflows. This work highlights the potential of SDA as a scalable and practical solution for generating synthetic seismic datasets that closely resemble real-world observations.

Introduction

Seismic data are foundational to subsurface exploration and reservoir characterization. Through seismic interpretation, geoscientists delineate structural and stratigraphic features, assess hydrocarbon potential, and reduce geological uncertainty in development planning (Brown, 2011; Sheriff and Geldart, 1995). However, seismic data are frequently constrained by sparse acquisition geometry, ambient noise, limited resolution, and incomplete well control. These factors introduce ambiguities that make interpretation inherently uncertain and highly dependent on the interpreter's experience and assumptions (Bond et al., 2007).

Recent developments in data-driven methods—particularly machine learning (ML) and deep learning (DL)—have shown promise for automating seismic interpretation tasks such as fault detection, horizon tracking, and facies classification (Waldeland et al., 2018; Di et al., 2018). These models offer substantial efficiency gains and can improve consistency across large 3D seismic volumes. However, their performance critically depends on access to large, diverse, and accurately labeled datasets. Field datasets are often geographically limited, suffer from acquisition artifacts, and are sparsely labeled due to the labor-intensive nature of manual labeling (Wu et al., 2020).

To address this data limitation, the geoscience community increasingly turns to synthetic data generation and augmentation. Among these efforts, seismic data augmentation (SDA) has emerged as a promising approach for generating realistic datasets that emulate the structural and geophysical variability seen in the field (Pham and Fomel, 2023). Unlike purely observational datasets, augmented seismic data can be tailored to explore specific geologic scenarios, control noise levels, and provide access to synthetic ground truth—all of which are critical for testing and validation.

Despite their practical value, conventional SDA techniques—such as geometric transformations, amplitude scaling, or wavelet-based modeling—offer only limited realism. While these methods introduce variability through randomization, they fail to capture complex physical behaviors such as stochastic scattering, multipathing, and internal multiples—phenomena that are vital for interpreting structurally and lithologically complex environments (Zhu et al., 2020; Wu et al., 2020; Wang et al., 2022). Moreover, these techniques generally lack mechanisms to preserve structural coherence or introduce statistically consistent uncertainty, limiting their utility in workflows requiring physical plausibility and geological realism (Pham and Fomel, 2023; Guitton, 2005).

In this work, we present SDA —a physics-informed data augmentation framework that integrates structural modeling, geostatistics, and seismic wavefield simulation. The method consists of three key components. First, Image-Guided Interpolation (IGI) constructs deterministic velocity models by interpolating sonic logs in alignment with seismic image structures. Second, Sequential Gaussian Simulation (SGS) introduces small-scale heterogeneity based on geostatistical parameters, capturing realistic spatial variability and model uncertainty. Third, reflectivity-based forward modeling simulates wavefields that include primary reflections, internal multiples, or both—yielding synthetic seismic volumes with realistic amplitude behavior.

We demonstrate the SDA framework using a public field dataset. Although no ML model is trained in this study, the augmented volumes provide high-quality examples for interpretation and serve as potential inputs for future data-driven research. The results show that SDA generates seismic data that are geologically plausible, statistically diverse, and physically consistent—offering a robust platform for advancing both manual and automated interpretation workflows.

Theory

This study introduces a seismic data augmentation framework aimed at synthesizing realistic seismic volumes by integrating geological structure, geostatistical variability, and wavefield physics. The method combines multiple sources of subsurface information—including post-stack seismic images, interpreted horizons, and well log data—to build geologically consistent and diverse synthetic datasets. The approach incorporates structural constraints from seismic images, spatial statistics from interpolated logs, and physically grounded wavefield modeling to simulate realistic seismic responses.

The complete workflow is illustrated in Figure 1. It begins with seismic images and well logs as primary inputs, which are used in the image-guided interpolation (IGI) step to construct a structurally conformable velocity model. From this model, a set of control points—referred to as hard data—is defined using three sources: well log measurements, velocity values at horizon locations extracted from the IGI-generated model, and additional randomly sampled points within the model volume. These hard data collectively serve to constrain the subsequent geostatistical modeling.

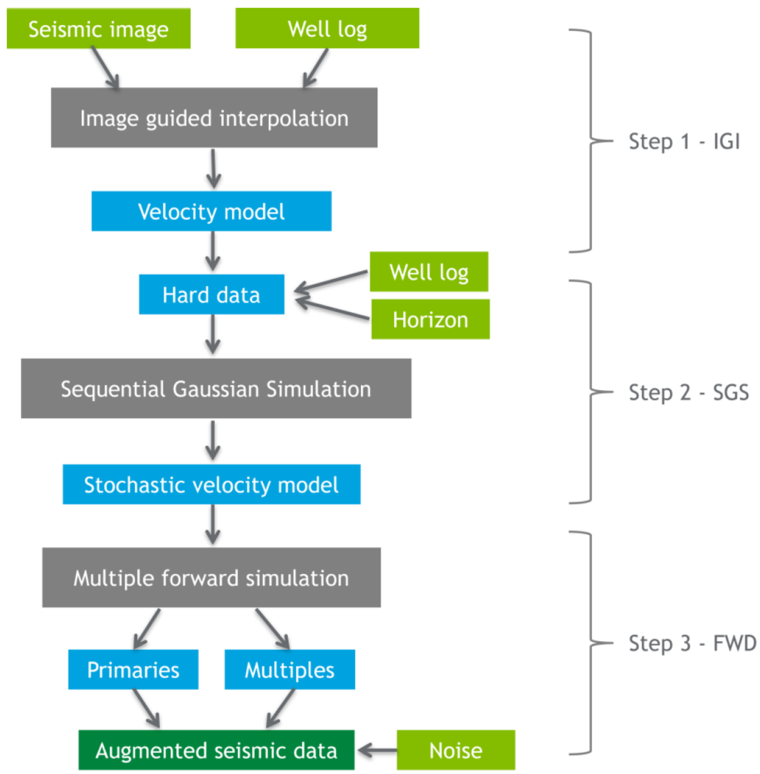


Figure 1—Workflow of the seismic augmentation method. Light green boxes (seismic image, well log, horizon, wavelet, and noise) are the inputs, dark green box (augmented seismic data) is the final output. Light blue boxes are intermediate results; gray boxes represent the main processing steps: IGI, SGS, and forward simulation.

Next, the sequential Gaussian simulation (SGS) step uses the hard data as input and generates stochastic realizations that reproduce the statistical variability of the interpolated model while honoring the defined constraints. The result is a stochastic velocity model that better reflects subsurface heterogeneity.

In the final stage, the stochastic velocity model is passed into the reflectivity-based forward modeling engine, which simulates seismic responses including both primary and multiple reflections. These simulated components are then combined and optionally merged with random noise to produce the final augmented seismic dataset. The intermediate results—such as the primaries, multiples, and stochastic velocity model—can also be retained for use in auxiliary interpretation or machine learning workflows.

Prior to executing the workflow, a streamlined preprocessing module is applied to align all input data onto a unified seismic grid. This includes integrating well logs, deviation surveys, time-depth (T-Z) conversion files, seismic volumes, and user-formatted horizon data. The module ensures consistent spatial and temporal alignment across datasets, minimizing manual intervention and reducing the potential for human error.

Image-guided Interpolation (IGI): generating structure-aligned velocity models from well logs

Image-guided interpolation (IGI) is a structure-conforming interpolation technique that propagates sparse well log measurements—such as velocity or density—along seismic reflectors in a manner consistent with subsurface structure (Hale et al., 2010). Rather than relying on purely spatial proximity, IGI uses local structural orientations derived from a seismic image to guide the interpolation process. This approach helps preserve horizon alignment and fault boundaries in the interpolated property model.

Let $m(\boldsymbol{x})$ represent the model to be interpolated (e.g., velocity), where $\boldsymbol{x}=(x,y,z)$ denotes spatial coordinates. The goal is to solve the following anisotropic diffusion equation used for structure-aligned interpolation:

$$\nabla \cdot (D(\boldsymbol{x})\nabla m(\boldsymbol{x})) = 0, \tag{1}$$

where $D(\boldsymbol{x})$ is a symmetric, positive-definite diffusion tensor that encodes the local directional structure of the seismic image.

The diffusion tensor D is constructed using eigenvectors of the structure tensor computed from the seismic image $I(\boldsymbol{x})$. Let v_1, v_2, v_3 denote the eigenvectors corresponding to the dominant, intermediate, and least variation directions in the local neighborhood, and let λ_1, λ_2, λ_3 be the associated eigenvalues. The tensor is then defined as:

$$D = \lambda_1 v_1 v_1^T + \lambda_2 v_2 v_2^T + \lambda_3 v_3 v_3^T. \tag{2}$$

To emphasize interpolation along seismic structures while suppressing cross-structure diffusion, the eigenvalues are typically chosen such that $\lambda_1 \gg \lambda_2, \lambda_3$, promoting strong diffusion along reflectors. The resulting equation is solved numerically with sparse hard constraints $m(x_i) = m_i$ at well log locations x_i, where measurements are available. The interpolation fills the 3D domain with structurally consistent property values.

This makes IGI particularly effective for generating property models that are not only smooth but also geologically plausible, as they conform to the dominant reflector geometry. The method preserves both structural coherence and stratigraphic alignment, which are critical for the reliability of subsequent stochastic simulations and seismic forward modeling.

Sequential Gaussian Simulation (SGS): introducing stochastic heterogeneity into velocity models

While IGI produces a smooth velocity model that aligns with structural features, it lacks the fine-scale variability and uncertainty inherent in real geological formations. To capture these stochastic variations, we apply Sequential Gaussian Simulation (SGS), a widely used geostatistical method for generating multiple realizations of spatially correlated property fields (Isaaks and Srivastava, 1989).

Prior to performing SGS, we identify and prepare a set of hard data points that constrain the simulation. These hard data include: (1) well log measurements at borehole locations, (2) interpolated velocity values at user-interpreted horizon positions from the IGI model, and (3) additional randomly selected points from the IGI-generated model to improve spatial coverage. These constraints provide both structural and statistical guidance for the SGS process, ensuring that simulated realizations honor known data and reflect realistic spatial variability.

The SGS process consists of the following steps:

a. **Detrending**: The IGI velocity model $V(x, y, z)$ is decomposed into a large-scale trend $V^{\text{trend}}(x, y, z)$ and a residual component $R(x, y, z)$:

$$R(x, y, z) = V(x, y, z) - V^{trend}(x, y, z). \tag{3}$$

The trend is estimated by applying a 3D Gaussian smoothing filter to the IGI velocity field:

$$V^{trend}(x, y, z) = G(x, y, z) * V(x, y, z), \tag{4}$$

where $G(x, y, z)$ is a normalized Gaussian kernel and * denotes convolution. This step ensures that SGS simulates only the stationary residual field $R(x, y, z)$, isolating small-scale fluctuations from the smooth background trend.

b. **Variogram analysis:** Using the detrended hard data $R(x_i, y_i, z_i)$, we compute the experimental variogram $\gamma(h)$:

$$\gamma(h) = \frac{1}{2N(h)} \sum \left[R(x_i, y_i, z_i) - R(x_i + h_x, y_i + h_y, z_i + h_z) \right]^2, \tag{5}$$

where $N(h)$ is the number of pairs separated by lag vector $h=(h_x, h_y, h_z)$. A theoretical variogram model (e.g., spherical, exponential, Gaussian) is then fitted to the experimental variogram to capture spatial continuity. The chosen model is used to define spatial correlations in the kriging step of the simulation.

c. **Simulation**: SGS is performed on the residual field $R(x, y, z)$.

d. **Trend restoration:** The final velocity model $V'(x, y, z)$ is reconstructed by adding the simulated residuals to the original trend:

$$V'(x, y, z) = R(x, y, z) + V^{trend}(x, y, z) \tag{6}$$

This output preserves both the structural trend imposed by IGI and the local variability introduced by SGS.

Reflectivity Forward Modeling: Generating Primary and Internal Multiple Reflections

With the stochastic velocity models generated from SGS, we simulate synthetic seismic responses that account for both primary reflections and internal multiples using reflectivity-based forward modeling. This method, derived from the 1D convolutional model of seismic wave propagation, is computationally efficient and capable of isolating distinct wavefield components.

The reflectivity method constructs the reflection response of a stratified medium to incident down-going waves in a two-stage recursive process. In detail, given a 1-D N-layer velocity model $v(z_{(i)})$, $i=1,\ldots,N,$ where $z_{(i)}$ is the depth of the i-th reflector, the reflection and transmission coefficients at each interface for incident downgoing and upgoing waves are given by

$$r_d(z_i) = \frac{v(z_{i+1}) - v(z_i)}{v(z_{i+1}) + v(z_i)},$$
$$t_d(z_i) = 1 - r_d(z_i),$$
$$r_u(z_i) = -r_d(z_i),$$
$$t_u(z_i) = 1 - r_u(z_i),$$

where $r_{(d)}(z(i))$ and $t_{(d)}(z_{(i)})$ are reflection and transmission coefficients for incident downgoing waves, respectively; $r_{(u)}(z_{(i)})$ and $t_{(u)}(z_{(i)})$ are reflection and transmission coefficients for incident upgoing waves, respectively $\overline{N-1}$

$$R_D(z_{N-1}^-) = r_d(z_{N-1}) \tag{7}$$

where z_{N-1}^- indicates the depth just above the layer z_{N-1}.$R_D(z_{N-1}^-)$ is the reflection coefficient at the bottom layer $z_{(N-1)}$. Next, we add the phase terms and compute the reflection below the interface at z_{N-2}^+

$$R_D(z_{N-2}^+) = E_D^{N-1} R_D(z_{N-1}^-) E_U^{N-1}, \tag{8}$$

where z_{N-2}^+ indicates the depth just below the layer $z_{(N-2)}$, and the phase increments in transmission E_D^{N-1} and E_U^{N-1} within the layer N-1 depend on the frequency ω and velocity

$$E_D^{N-1} = E_U^{N-1} = e^{i\omega \frac{z_{N-1} - z_{N-2}}{v(z_{N-1})}}. \tag{9}$$

After incorporating the interface at $z_{(N-2)}$, we bring the reflection to the upper side of the same layer. Then compute the reflection above the interface at z_{N-2}^-

$$\begin{aligned} R_D\left(z_{N-2}^{-}\right) &= r_d\left(z_{N-2}\right)+t_d\left(z_{N-2}\right)R_D\left(z_{N-2}^{+}\right)\left(1-r_u\left(z_{N-2}\right)R_D\left(z_{N-2}^{+}\right)\right)^{-1}t_u\left(z_{N-2}\right) \\ &= r_{N-2}+\left(1-r_{N-2}^{2}\right)R_D\left(z_{N-2}^{+}\right)\left(1+r_{N-2}R_D\left(z_{N-2}^{+}\right)\right)^{-1}. \end{aligned} \quad (10)$$

Note that the term $\left(1-r_u\left(Z_{N-2}\right)R_D\left(z_{N-2}^{+}\right)\right)^{-1}$ in equation (10) represents the cumulative effect of a sequence of internal reverberations, containing primaries and infinite orders of internal multiples. A Taylor series expansion of this term explicitly separate primary reflection, first-order multiples, and each high-order multiples explicitly.

This modeling scheme supports three output modes:

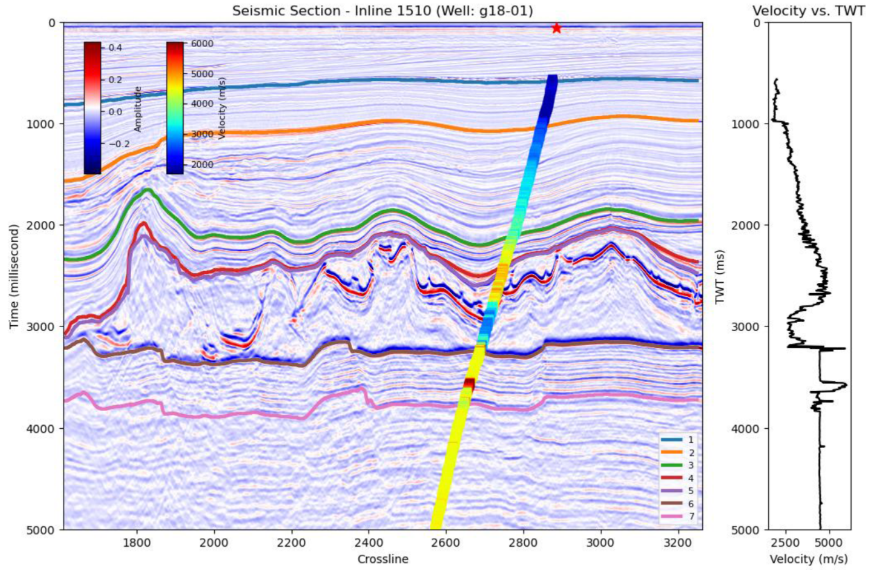
a. Primary-only: direct reflections from impedance contrasts without internal reverberations.
b. Multiple-only: energy paths involving two or more internal reflections.
c. Full wavefield: superposition of primary and multiple responses.

Because the reflectivity modeling is based on the layered velocity model from SGS, it inherently incorporates both large-scale structure and small-scale heterogeneity. This enables the generation of realistic seismic responses suitable for training machine learning models or benchmarking interpretation workflows.

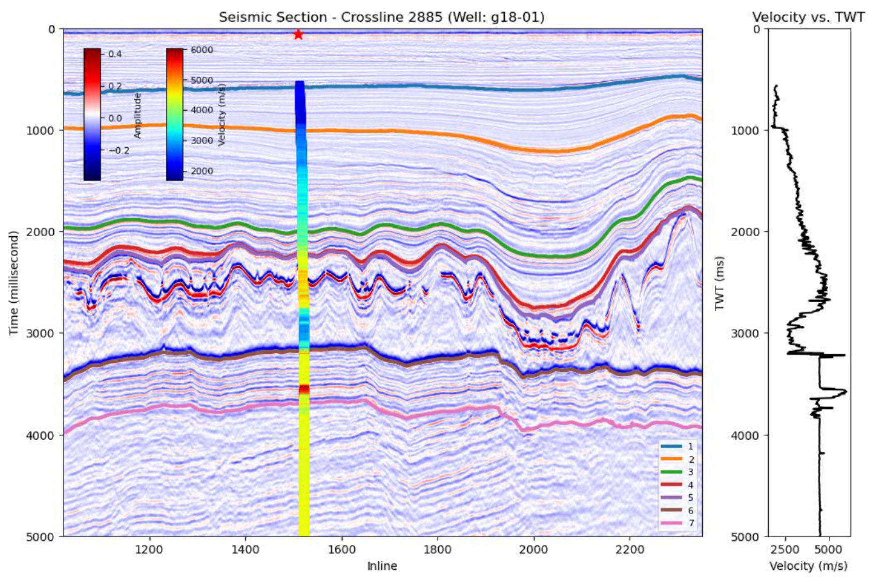
Field Data Test

The following example illustrates the application of SDA to a public 3D field dataset from the Netherlands. This case study demonstrates how the system processes input data, generates high-resolution velocity models, introduces stochastic uncertainties, and simulates realistic seismic wavefields.

Data preprocessing step involves extracting and standardizing various input data including prestack time migration image, well logs and interpreted horizons, ensuring proper alignment for subsequent modeling. The seismic section visualizations in Figure 2 present inline (2a) and crossline (2b) images corresponding to the wellhead locations, highlighting the spatial integration of seismic data, well trajectories, and geological horizons. The amplitude of the seismic data is represented as an intensity map, with well trajectories overlaid to illustrate the spatial correlation between well log data and the seismic grid. Velocity data along the well trajectories are displayed with marker colors representing velocity magnitude. Red stars denote wellhead locations. Geological horizons, aligned with the seismic grid, are shown as continuous lines to emphasize the integration of interpreted subsurface structures with seismic data. On the right of each image, well log velocity is plotted using wiggle traces, providing an additional view of the velocity distribution along the well trajectory.



(a)



(b)

Figure 2—Inline (2a) and crossline (2b) seismic sections at the wellhead location, displaying seismic amplitude, overlaid well trajectories, velocity data, wellhead markers, geological horizons, and well log velocity wiggle traces.

Figure 3 provides a comparative visualization of the seismic image sections, and the velocity model generated using Image-Guided Interpolation (IGI). The top row displays the inline section, and the bottom row represents the crossline section, with the seismic amplitude shown on the left and the corresponding IGI-generated velocity model on the right. In both the inline and crossline views, velocity data along the well log trajectory is overlaid on the seismic image sections. This overlay highlights the spatial correlation between the well log-derived velocities and the subsurface reflectivity patterns in the seismic data. The IGI-generated velocity model, shown on the right in both rows, captures large-scale geological structures with smooth and consistent variations. By incorporating well log data, the velocity model ensures alignment with observed seismic features, maintaining structural fidelity and enhancing geological realism.

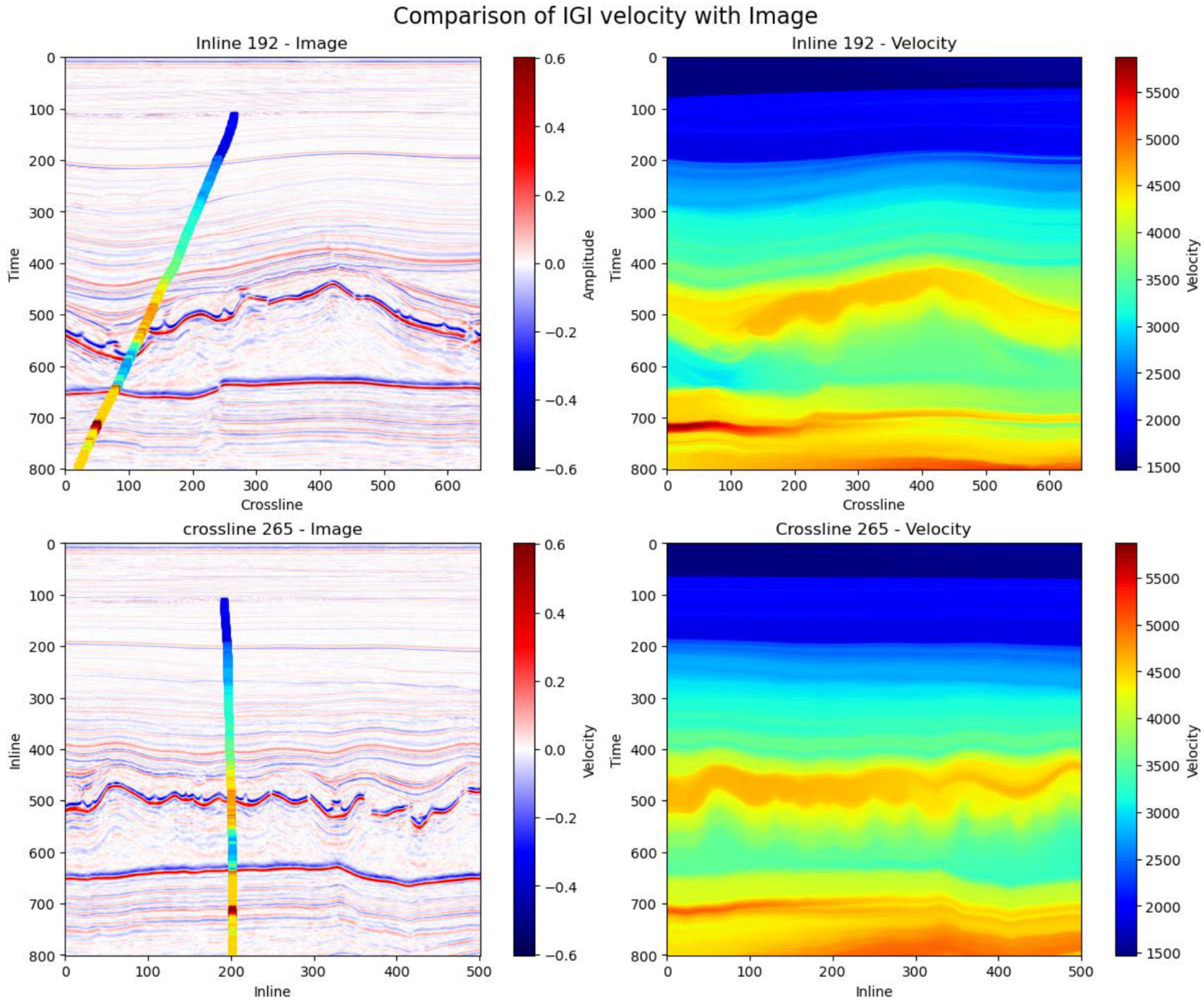


Figure 3—Comparison of seismic image sections (left) and IGI-generated velocity models (right), with well log velocity data overlaid on the seismic images for inline 192 (top) and crossline 265 (bottom).

Figure 4 highlights the analysis and results of the SGS process at time slice 154, focusing on the experimental variogram and the comparison between the IGI result, selected hard data, and the SGS result. The top figure presents the isotropic experimental variogram for time slice 154. The experimental variogram (green dots) captures the spatial correlation of velocity values, revealing how geological properties vary with distance. Two fitted variogram models, exponential (red line) and spherical (blue line), accompany the experimental data. These models are fundamental to the SGS process, ensuring that the simulated geological variations are both geologically realistic and statistically consistent with observed spatial trends. The bottom figure compares the IGI velocity model, selected hard data points, and the SGS results for the same time slice. The IGI velocity model (left panel) exhibits smooth, large-scale structural trends, whereas the SGS velocity model (right panel) incorporates stochastic variations, adding finer-scale geological realism. The hard data, shown in the center panel, consists of well log data, interpreted horizons, and randomly selected points from the IGI velocity model. These hard data points serve as essential constraints in the SGS process, ensuring alignment with known geological features while introducing realistic variability. The SGS result demonstrates the successful integration of large-scale structural trends from the IGI model with small-scale variations guided by the variogram, delivering a geologically robust velocity model.

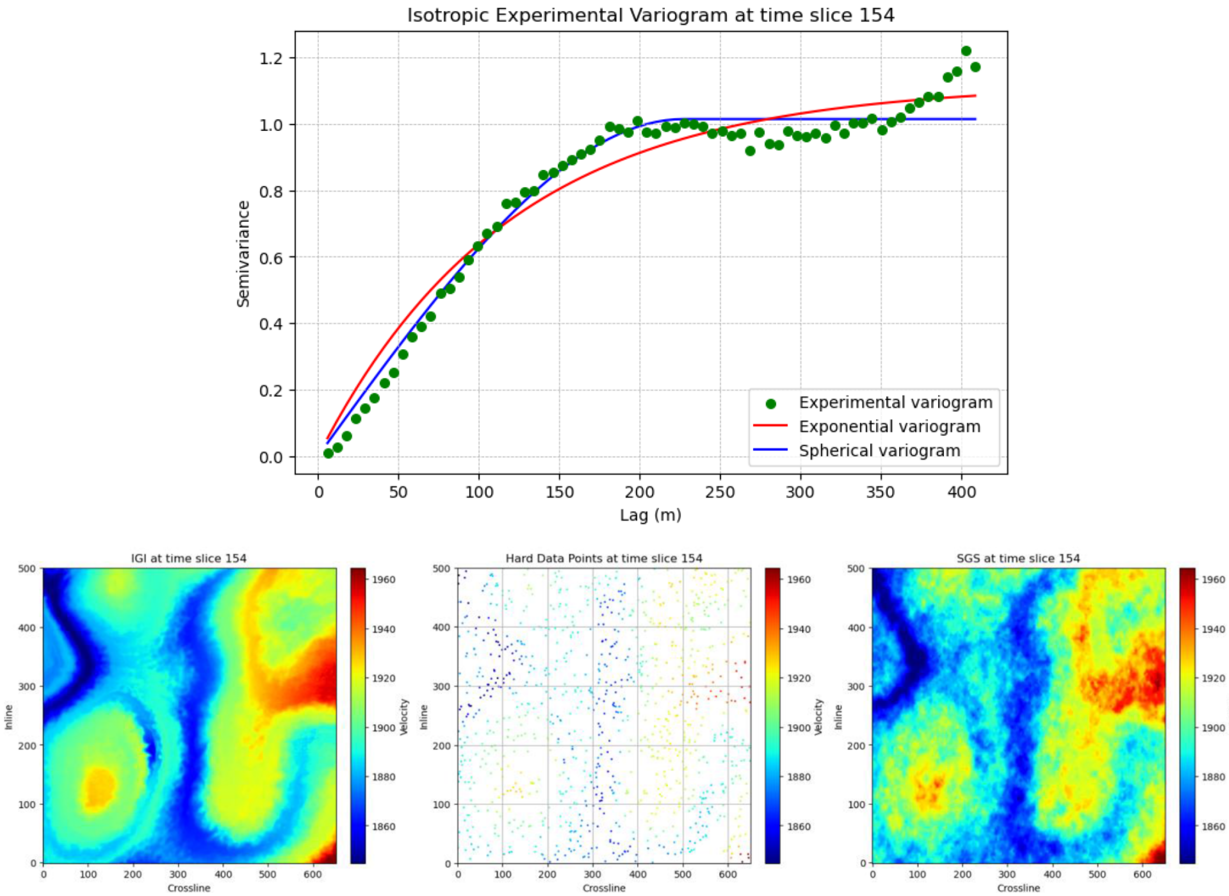


Figure 4—Analysis and results of the SGS process at time slice 154, including the isotropic experimental variogram with fitted models (top) and the comparison of the IGI velocity model, selected hard data points, and SGS result (bottom).

Figure 5 compares the IGI-generated velocity model (left) with the SGS-enhanced velocity model (right) across inline, crossline, and time slice views. The comparison demonstrates how SGS preserves the structural trends established by IGI while introducing small-scale stochastic variations that enhance geological realism. In particular, the bottom row showcases time slices for IGI (left) and SGS (right), with the SGS time slice displaying increased local variability and finer-scale geological details compared to the smoother IGI result. This figure underscores how SGS enriches the velocity model with realistic subsurface heterogeneities while maintaining alignment with the large-scale structural trends of the IGI model.

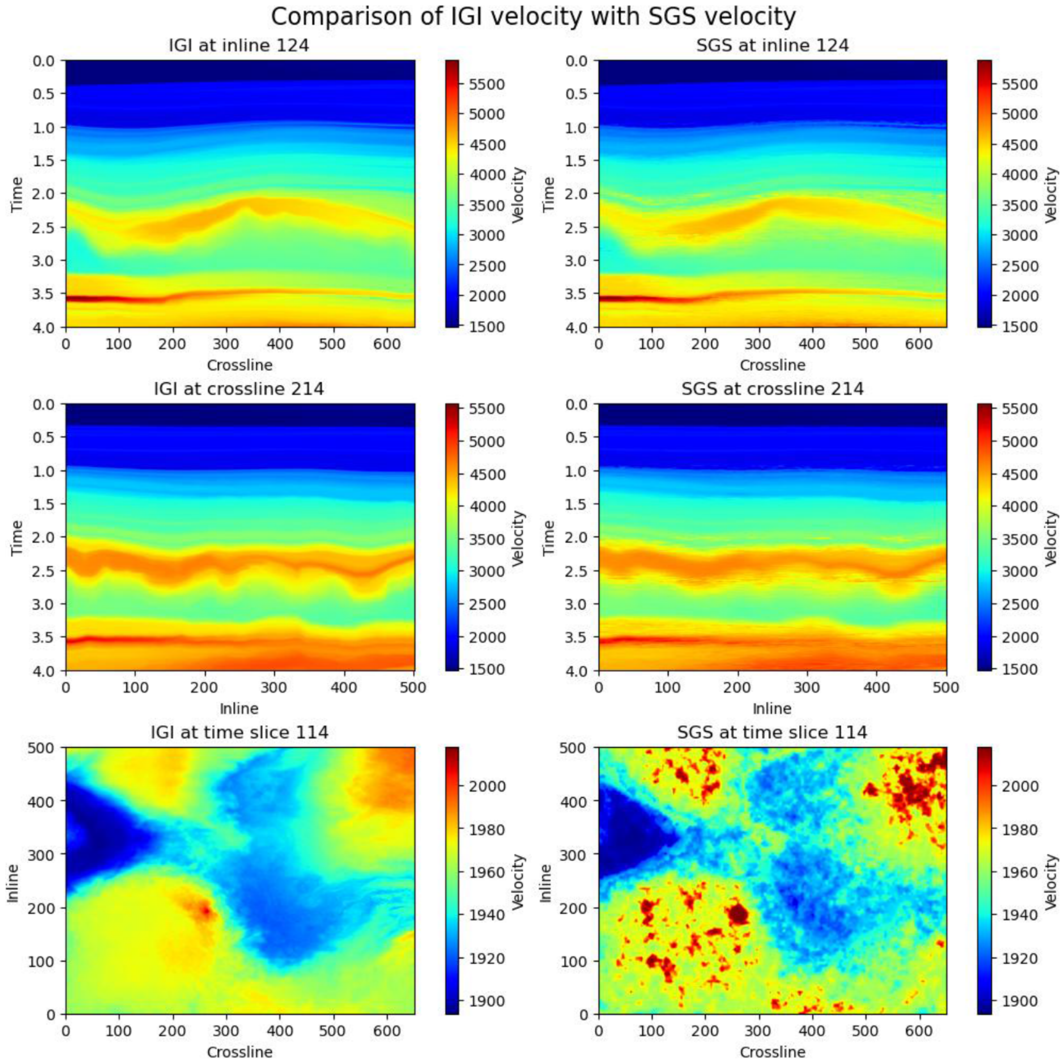


Figure 5—Comparison of IGI velocity model (left) and SGS-enhanced velocity model (right)

In the multiple forward modeling process, we source wavelet is first estimated from the seismic image and IGI velocity cubes. Figure 6 illustrates the extracted 3D wavelet and its spectral characteristics after phase correction. The left panel displays the wavelet in the time domain, capturing the temporal characteristics of the source wavelet used in the seismic acquisition process. The blue curve represents the wavelet directly extracted from the PSTM image using the multi-channel Wiener filter, while the red curve shows the save wavelet after applying phase correction to achieve a zero-phase response. The middle and right panels present the corresponding amplitude spectrum and phase spectrum in the frequency domain, comparing the minimum-phase wavelet (blue) and the corrected zero-phase wavelet (red). This extracted wavelet is derived by applying a multi-channel Wiener filter to the seismic image cube and the IGI velocity data around the well trajectory. The post-processed zero-phase wavelet serves as an essential input for forward modeling, ensuring that the simulated seismic wavefields closely replicate the spectral characteristics of the real data, thereby enhancing the realism and fidelity of the forward modeling process.

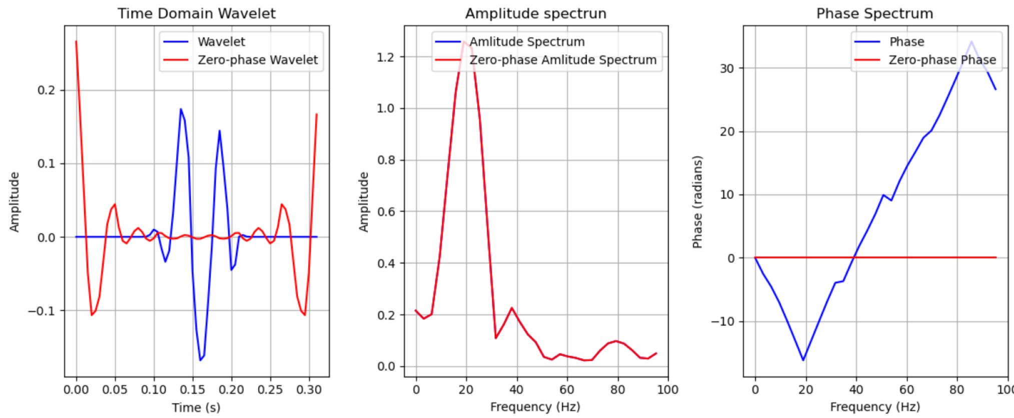


Figure 6—Extracted 3D wavelet in the time domain (left), amplitude spectrum (middle) and phase spectrum (right).

Figure 7 presents a detailed comparison of the full wavefield, primary reflections, and internal multiples generated using the IGI velocity model. Inline, crossline, and time slice views demonstrate the differences between these components. The full wavefield captures the comprehensive seismic response, while the primary reflections isolate the direct geological interfaces. Internal multiples, depicted separately, showcase wavefield complexity and highlight regions with multiple reflection events, crucial for enhancing seismic interpretation.

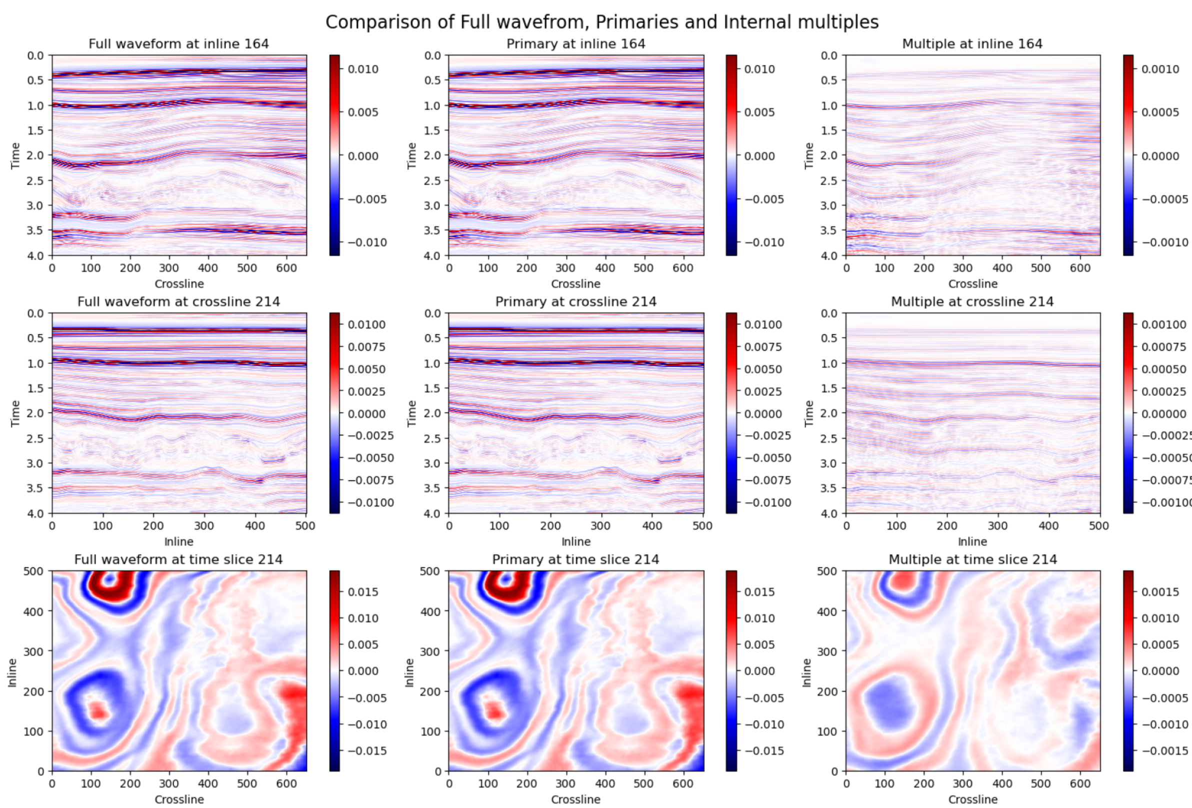


Figure 7—Comparison of the full wavefield, primary reflections, and internal multiples across inline, crossline, and time slice views using the IGI velocity model.

Figure 8 contrasts the IGI velocity model, synthetic seismic data, and the pre-stack time migration (PSTM) image across inline, crossline, and time slice views. The IGI velocity model provides a foundational representation of subsurface structures, while the synthetic seismic data, generated using forward modeling, aligns closely with the PSTM image, validating the accuracy of the augmented seismic workflow. This comparison emphasizes the consistency between the synthetic data and real seismic images, showcasing how the velocity model guides realistic seismic wavefield generation and enhances interpretation fidelity.

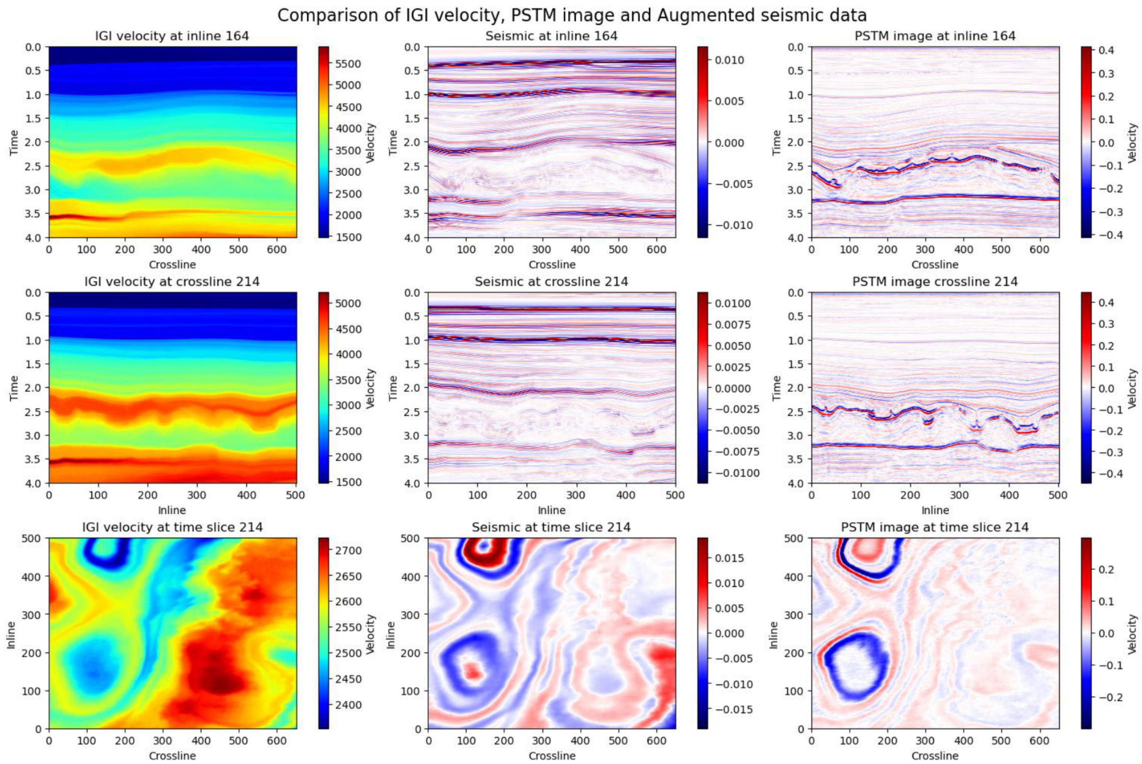


Figure 8—Comparison of the IGI velocity model, synthetic seismic data, and the PSTM image across inline, crossline, and time slice views, highlighting the consistency between modeled and real seismic data.

Using the SGS velocity model, we can generate similar figures to compare the full wavefield, primary reflections, and internal multiples, as well as to evaluate the consistency between the SGS-generated synthetic seismic data and the PSTM image. These figures would highlight the impact of small-scale stochastic variations introduced by SGS on the seismic wavefields and their alignment with observed seismic data.

Conclusion

This study introduces a modular and extensible framework for seismic data augmentation that combines structural interpolation, geostatistical simulation, and wavefield modeling. The method consists of three main components. First, seismic image-guided interpolation (IGI) transforms sparse well log data into a continuous velocity model that aligns with structural features in the seismic image. This step ensures that the large-scale geological framework is honored. Second, sequential Gaussian simulation (SGS) is applied to introduce small-scale spatial variability. Hard data, including well log values, interpreted horizon points, and random seed points, are used to condition the simulation. It incorporates a detrending process where smooth trends are removed for variogram modeling to isolate small-scale variability more effectively and restored afterward to retain large-scale geological structure in the final simulation. The result is a suite of velocity models that capture realistic subsurface heterogeneity. Third, reflectivity-based forward modeling simulates seismic responses using the generated velocity models. This process supports the isolation of primary reflections, internal multiples, or a combined wavefield, providing physically consistent datasets that reflect complex wave propagation phenomena observed in field data. Application of the method to a public 3D dataset demonstrates its ability to produce geologically plausible and physically meaningful synthetic seismic volumes. The results show structural alignment with the input image, statistical variability consistent with field data, and realistic wavefield characteristics.

The proposed approach offers a valuable tool for generating synthetic data in support of seismic interpretation and geophysical analysis. Its flexibility allows adaptation to different geological settings, survey geometries, and data types. Future work will explore the integration of these augmented datasets into deep learning workflows for supervised model training, domain generalization, and robustness evaluation in seismic interpretation tasks such as fault detection, horizon tracking, and lithology classification.

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